



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Falcon 9 First Stage Landing Prediction

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Methods**

- Collected 90 Falcon 9 launch records from the SpaceX v4 REST API and scraped the Wikipedia launch table with BeautifulSoup.
- Filled missing payload masses, standardised types, and collapsed eight landing-outcome strings into one binary Class label.
- Explored the data with six Matplotlib and Seaborn charts and ten SQL queries.
- Built a Folium map and a Plotly Dash dashboard for interactive analysis.
- Tuned four classifiers with GridSearchCV over 10-fold cross-validation and scored them on 18 held-out launches.

- **Results**

- Landing success rose from 0% before 2013 to 84% in 2020. Flight number is the strongest single predictor in the dataset.
- Orbit and payload mass separate outcomes more sharply than launch site does, though all three are correlated with each other.
- KSC LC-39A has the highest success ratio at 76.9%, but it also opened late and flew newer boosters.
- The decision tree scored 94.44% on the test set; logistic regression, SVM and KNN each scored 83.33%.
- The confusion matrix shows no false negatives and one false positive, so the model leans optimistic about recovery.

Introduction

- **Background**
 - SpaceX lists a Falcon 9 launch at \$62 million. Competing providers charge upwards of \$165 million.
 - Most of that gap comes from recovering and reusing the first stage, so whether the stage lands is really a question about what the launch costs.
 - A company bidding against SpaceX needs to know which launches recover their booster and which do not.
- **Questions this project answers**
 - Which launch parameters separate a recovered first stage from a lost one?
 - Do payload mass, orbit and launch site act independently, or are they tangled up with each other?
 - Which classification model predicts the landing outcome most accurately, and by how much?

Methodology

Methodology

Methodology

Data collection

SpaceX v4 REST API for flight number, launch site, payload mass, orbit, booster serial and landing outcome.

Wikipedia launch records table scraped with BeautifulSoup as a cross-check.

Data wrangling

Missing payload masses replaced with the column mean; Falcon 1 rows dropped, leaving 90 launches.

Eight landing-outcome strings collapsed into a single binary Class column.

Exploratory data analysis

Six charts covering flight number, payload mass, orbit type and year.

Ten SQL queries, from unique site names to a ranked count of landing outcomes.

Interactive visual analytics

Folium map with site markers, outcome-coloured clusters and measured distances to nearby features.

Plotly Dash app with a site dropdown and a payload range slider.

Predictive analysis

Logistic regression, SVM, decision tree and KNN, each tuned with GridSearchCV and scored on a held-out set.

Data Collection

- Two sources were combined into one table, keyed on flight number.
- **SpaceX REST API**
 - Returned launch records as JSON, normalised into a DataFrame. Supplied booster version, payload mass, orbit, launch site, flight number, grid fins, reused flag, legs, block and landing outcome.
- **Wikipedia web scraping**
 - An HTML table of Falcon 9 and Falcon Heavy launch records, parsed with BeautifulSoup. Supplied the booster landing outcome text and the customer field, which the API does not expose cleanly.
- Both were reduced to one row per launch, cleaned, and merged. Filtering to Falcon 9 left 90 launches.

Data Collection – SpaceX API

- **Process**
 - GET request to `api.spacexdata.com/v4/launches/past`
 - `response.json()` into `pd.json_normalize()`
 - Helper calls to `/rockets`, `/payloads`, `/launchpads` and `/cores` resolve ID references into names and values
 - Build a dictionary of the required columns, then a `DataFrame`
 - Filter to Falcon 9 only, leaving 90 launches
 - Replace missing `PayloadMass` with the column mean
 - Export to `dataset_part_1.csv`
- **GitHub:** `jupyter-labs-spacex-data-collection-api.ipynb`
- <https://github.com/akhileshm0507/IBM-Data-Science-Capstone-Project>

Data Collection - Scraping

- **Process**
 - `requests.get()` on the static Wikipedia snapshot of "List of Falcon 9 and Falcon Heavy launches"
 - `BeautifulSoup(response.text, "html.parser")`
 - `soup.find_all("table")` to locate the launch records table
 - `extract_column_from_header()` parses the `<th>` elements into column names
 - Iterate the `<tr>` rows, parsing each `<td>` into the launch dictionary
 - Convert the dictionary to a `DataFrame` and export to `spacex_web_scraped.csv`
- **GitHub: jupyter-labs-webscraping.ipynb**
- <https://github.com/akhileshm0507/IBM-Data-Science-Capstone-Project>

Data Wrangling

- **Load dataset → count outcomes → classify → create Class label → export**
 - Counted launches per site (CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E) to check the distribution.
 - Counted each orbit type: GTO, ISS, VLEO, LEO, SSO, MEO, HEO, ES-L1, SO, GEO and PO.
 - Counted the landing outcome strings, which produced eight distinct outcomes.
 - Defined the bad outcomes: False ASDS, False Ocean, False RTLS, None ASDS and None None.
 - Created the binary label Class — 1 where the landing succeeded, 0 otherwise.
 - Took the mean of Class to confirm the overall landing success rate.
- **GitHub: labs-jupyter-spacex-Data wrangling.ipynb**
- <https://github.com/akhileshm0507/IBM-Data-Science-Capstone-Project>

EDA with Data Visualization

- Six charts, each aimed at one question about what separates a recovered booster from a lost one. Colour marks the landing outcome throughout.
 - Flight Number vs. Launch Site (scatter) — which sites were active in which era, and how outcomes track experience.
 - Payload Mass vs. Launch Site (scatter) — whether particular pads handle particular mass ranges.
 - Success Rate by Orbit Type (bar) — ranks the eleven orbits by landing success rate.
 - Flight Number vs. Orbit Type (scatter) — separates a real orbit effect from the effect of flying later.
 - Payload Mass vs. Orbit Type (scatter) — isolates mass from orbit instead of confounding the two.
 - Annual Success Rate (line) — landing reliability year by year from 2010 to 2020.
- Features were then one-hot encoded with `pd.get_dummies()` on Orbit, LaunchSite, LandingPad and Serial, and cast to float64.
- GitHub: `jupyter-labs-eda-dataviz.ipynb` — <https://github.com/akhileshm0507/IBM-Data-Science-Capstone-Project>

EDA with SQL

- Unique launch site names (DISTINCT).
- Five records where the site name begins with "CCA" (LIKE, LIMIT).
- Total payload mass carried for NASA (CRS) (SUM).
- Average payload mass for booster version F9 v1.1 (AVG).
- Date of the first successful ground pad landing (MIN).
- Boosters that landed on a drone ship carrying between 4,000 and 6,000 kg (BETWEEN).
- Count of launches by mission outcome (GROUP BY).
- Booster versions that carried the maximum payload mass (nested SELECT MAX).
- Every 2015 record, with the month name rebuilt from substr(Date, 6, 2).
- Landing outcomes between 2010-06-04 and 2017-03-20, counted and ranked descending.

- **GitHub: jupyter-labs-eda-sql-coursera_sqlite.ipynb**
- <https://github.com/akhilesh0507/IBM-Data-Science-Case-Study-Project>

Build an Interactive Map with Folium

- **Map objects added**
 - Circle and Marker at NASA Johnson Space Center as the opening view.
 - A 1,000 m circle and a labelled marker at each of the four launch sites, which is what makes the coastal, low-latitude pattern visible.
 - A MarkerCluster holding one marker per launch, green for success and red for failure. Clustering was necessary because many launches share the same coordinates.
 - MousePosition, to read off the latitude and longitude of nearby features.
 - PolyLines from a launch site to the nearest coastline, railway, highway and city, each labelled with its Haversine distance.
- The map tests whether pads are sited for safety and logistics: close to water and transport, clear of towns.
- **GitHub: `lab_jupyter_launch_site_location.ipynb`**
- <https://github.com/akhileshm0507/IBM-Data-Science-Capstone-Project>

Build a Dashboard with Plotly Dash

- **Components**

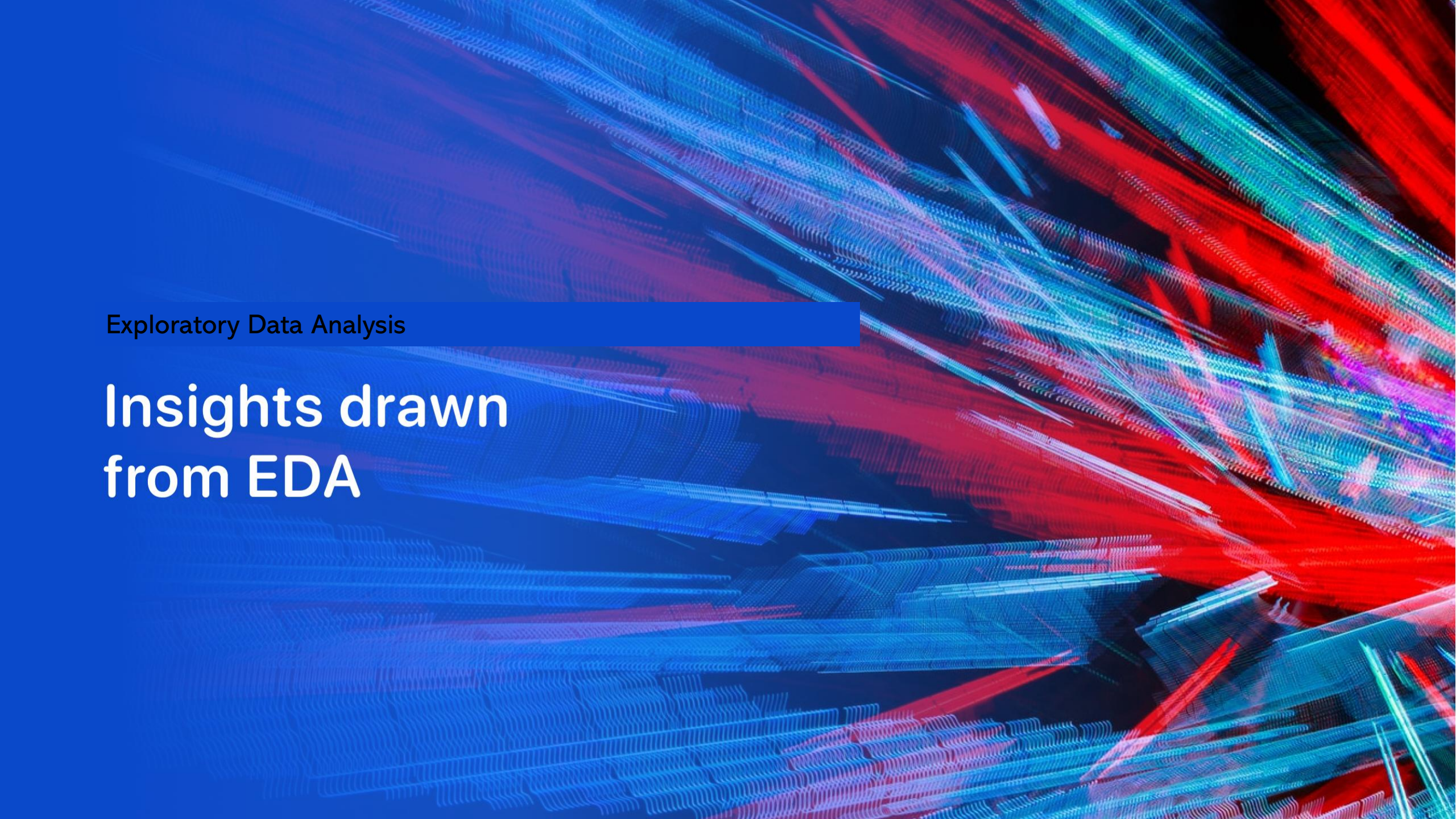
- A dropdown (dcc.Dropdown) for All Sites or one individual pad.
 - A pie chart (px.pie) showing the share of successful landings across sites, or success against failure for one selected site.
 - A range slider (dcc.RangeSlider) for payload mass from 0 to 10,000 kg in 1,000 kg steps.
 - A scatter plot (px.scatter) of payload mass against landing outcome, coloured by booster version.
-
- Site and payload are correlated in this data, so the dropdown and slider are the only way to look at one without the other. Colouring by booster version keeps the generation effect visible inside any payload window.
 - **GitHub: `spacex_dash_app.py`**
 - <https://github.com/akhileshm0507/IBM-Data-Science-Capstone-Project>

Predictive Analysis (Classification)

- **Load data → build Y from Class → standardise X → 80/20 split → GridSearchCV → score on test set → confusion matrix**
 - Y taken from the Class column; X standardised with StandardScaler so no feature dominates by scale.
 - train_test_split with random_state=2 and test_size=0.2, leaving 18 launches held out.
 - A parameter grid per algorithm, searched with GridSearchCV over 10-fold cross-validation.
 - Tuned C and kernel for SVM, max_depth and criterion for the decision tree, n_neighbors and p for KNN.
 - Each tuned model scored on the held-out launches, then plotted as a confusion matrix.
- Models built: logistic regression, support vector machine, decision tree and K-nearest neighbours.
- **GitHub: SpaceX_Machine_Learning_Prediction_Part_5.ipynb**
- <https://github.com/akhileshm0507/IBM-Data-Science-Capstone-Project>

Results

- Exploratory data analysis results
- Interactive analytics in screenshots
- Predictive analysis results

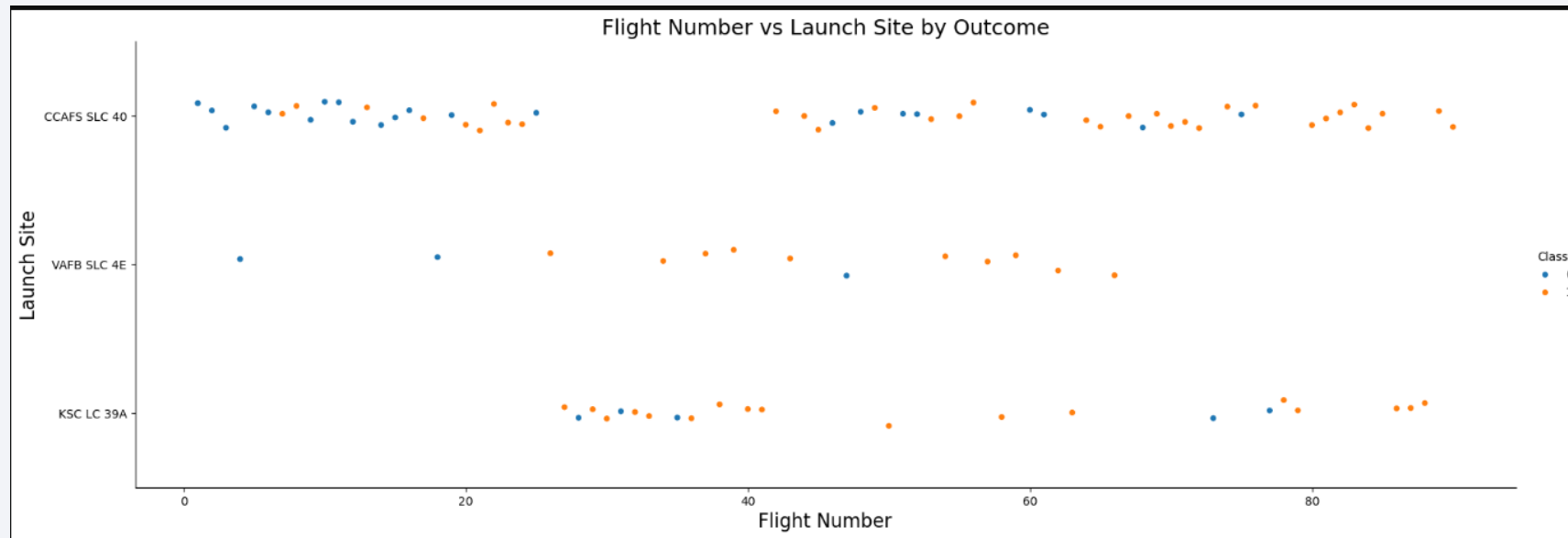
The background of the slide is an abstract composition of numerous thin, overlapping lines and streaks in shades of blue and red, creating a sense of motion and data. A solid blue rectangular block is positioned in the upper left area, serving as a backdrop for the text.

Exploratory Data Analysis

Insights drawn from EDA

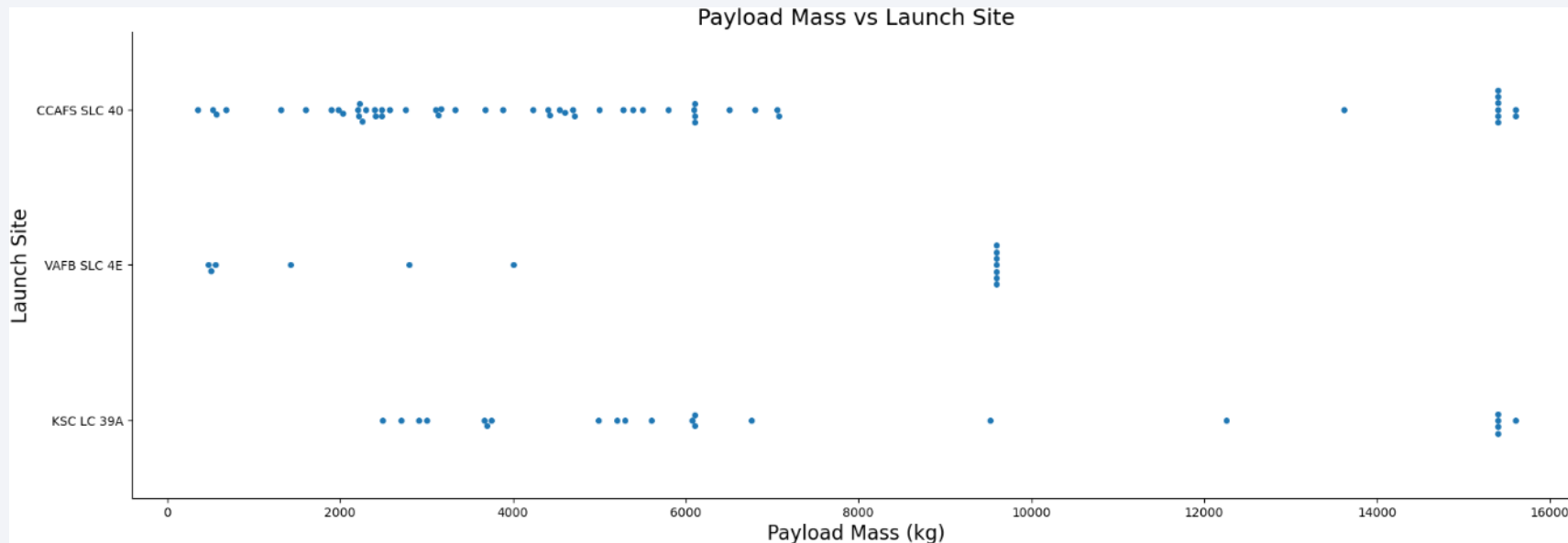
Flight Number vs. Launch Site

- **Mixed outcomes at every site.**
 - CCAFS SLC 40 and KSC LC 39A both show successes (orange) and failures (blue), so the pad on its own does not decide the outcome.
- **Early flights fail, later flights land.**
 - Nearly every launch below flight 30 is blue, and orange takes over past flight 40.
 - The same shift appears at all three sites, which points at accumulated experience rather than location.



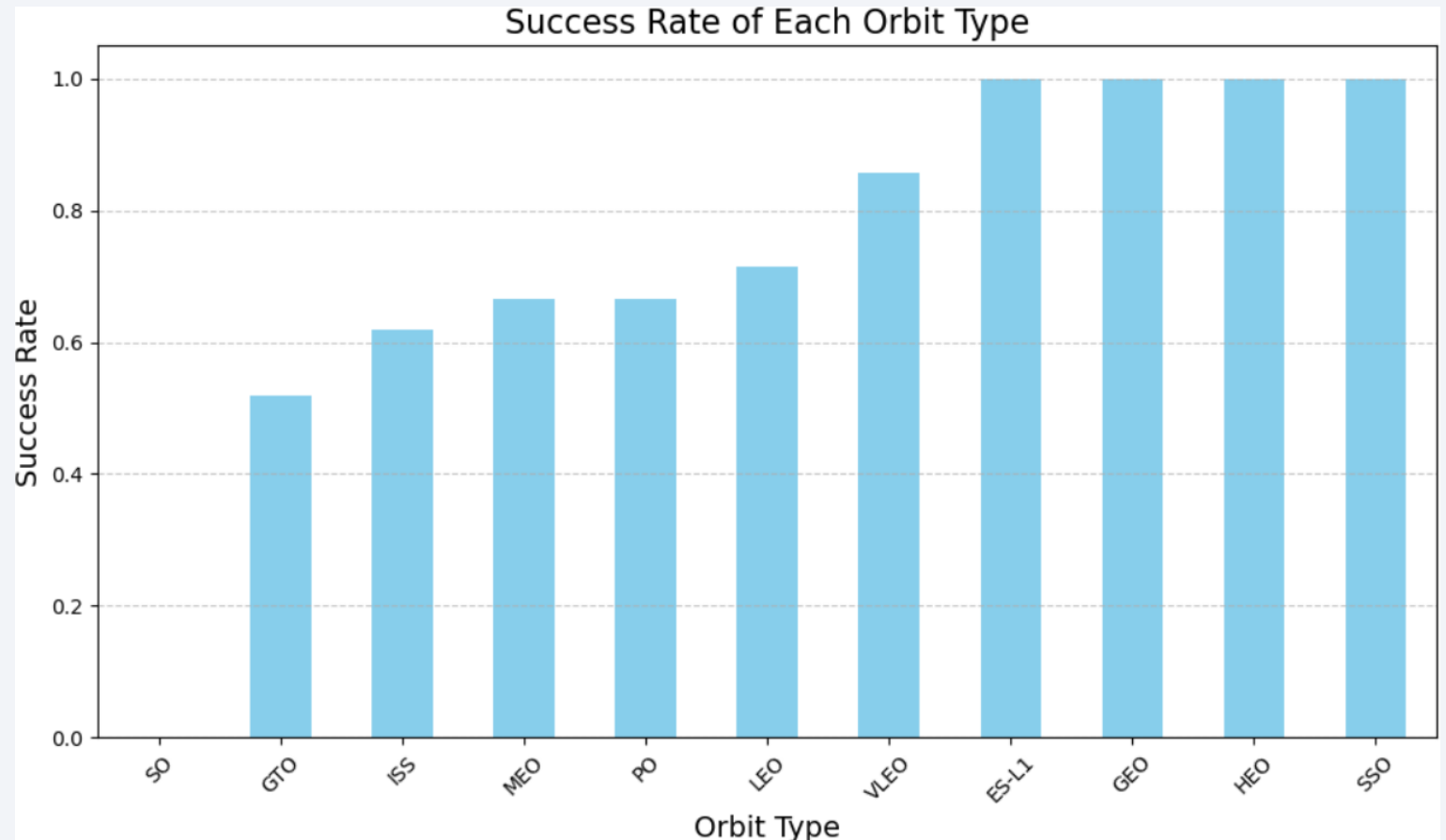
Payload vs. Launch Site

- **Payload distribution.**
 - CCAFS SLC 40 and KSC LC 39A both carry payloads across the full range, up to 15,600 kg.
 - VAFB SLC 4E tops out near 9,600 kg and flies the fewest launches.
- **Heavy launches cluster.**
 - Every point above 15,000 kg sits at one of the two Florida pads, so payload mass and launch site are not independent.



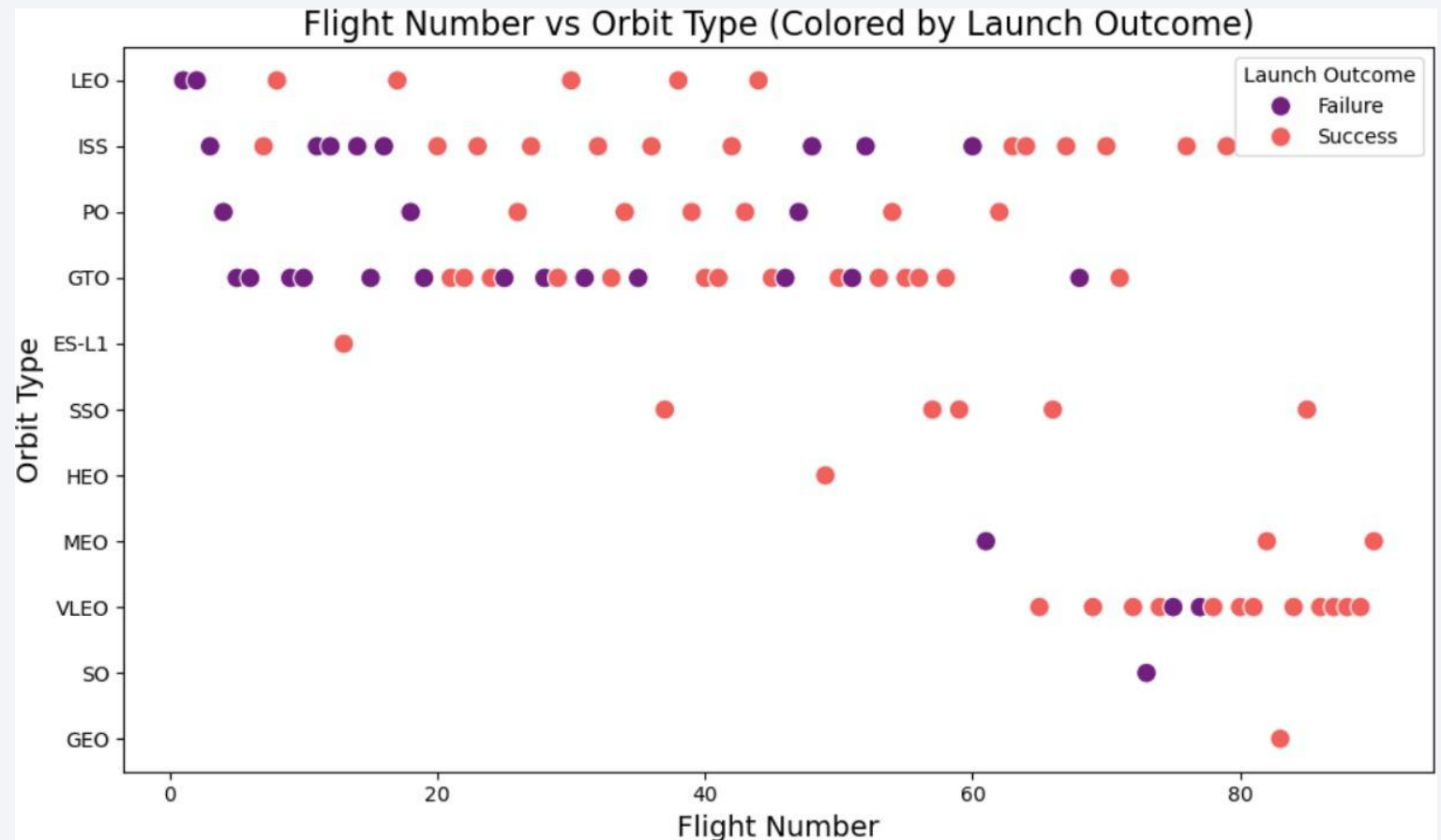
Success Rate vs. Orbit Type

- **Perfect records.**
 - ES-L1, GEO, HEO and SSO all sit at 1.0, but each rests on a handful of launches, so the top of the ranking is thin.
- **Weakest orbits.**
 - SO sits at 0 from a single launch.
 - GTO is the lowest of the well-sampled orbits at about 0.52.
 - VLEO reaches 0.86 across a larger sample, which makes it the most convincing high performer here.



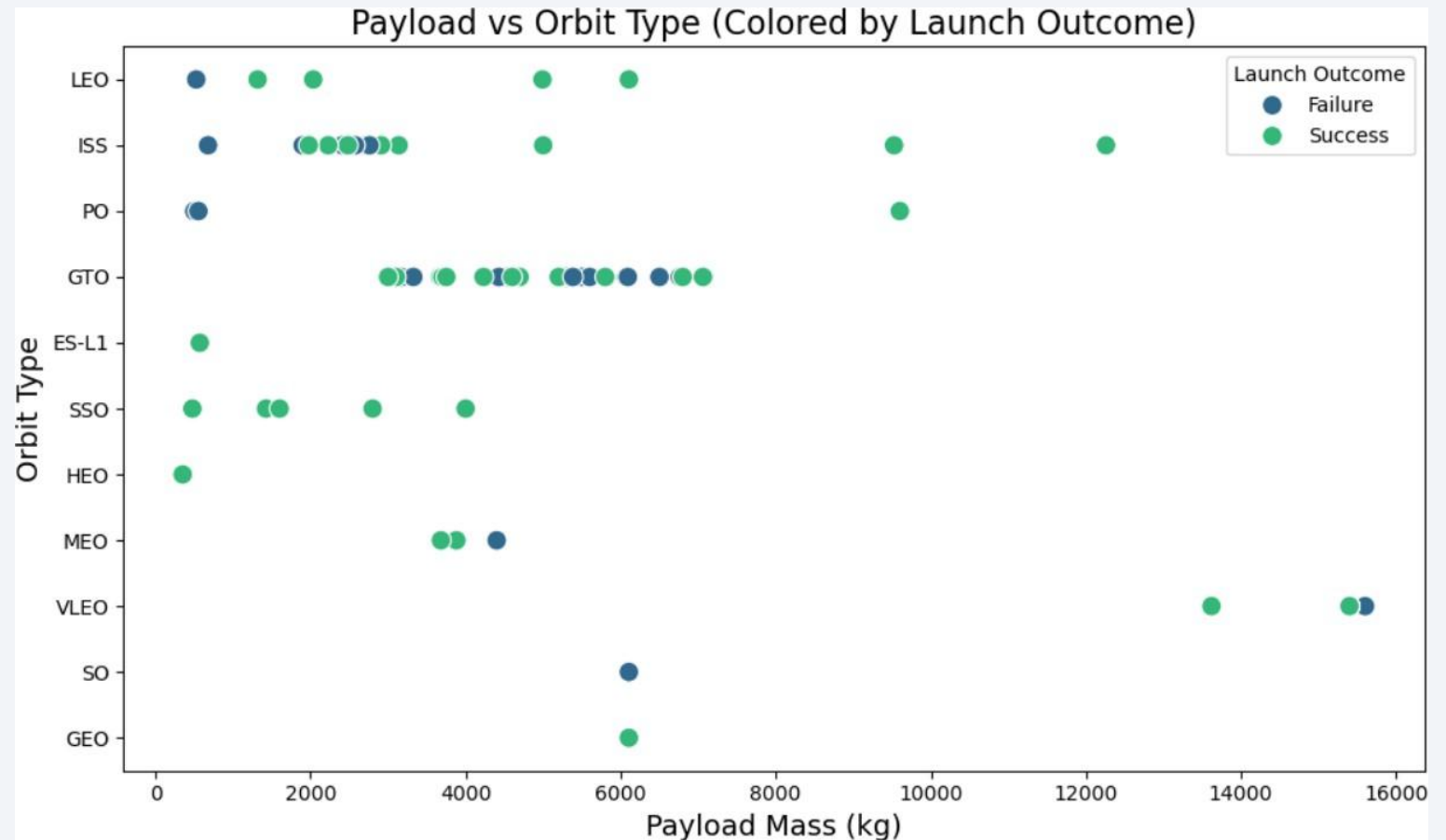
Flight Number vs. Orbit Type

- **Success climbs with flight number.**
 - Failures crowd the left of the plot and successes the right. Experience tracks the outcome more closely than any single orbit does.
- **Reading the orbits.**
 - GTO stays mixed across its whole range.
 - VLEO, SSO and GEO only appear past flight 60 and land almost every time, so their strong records come partly from when they were flown.



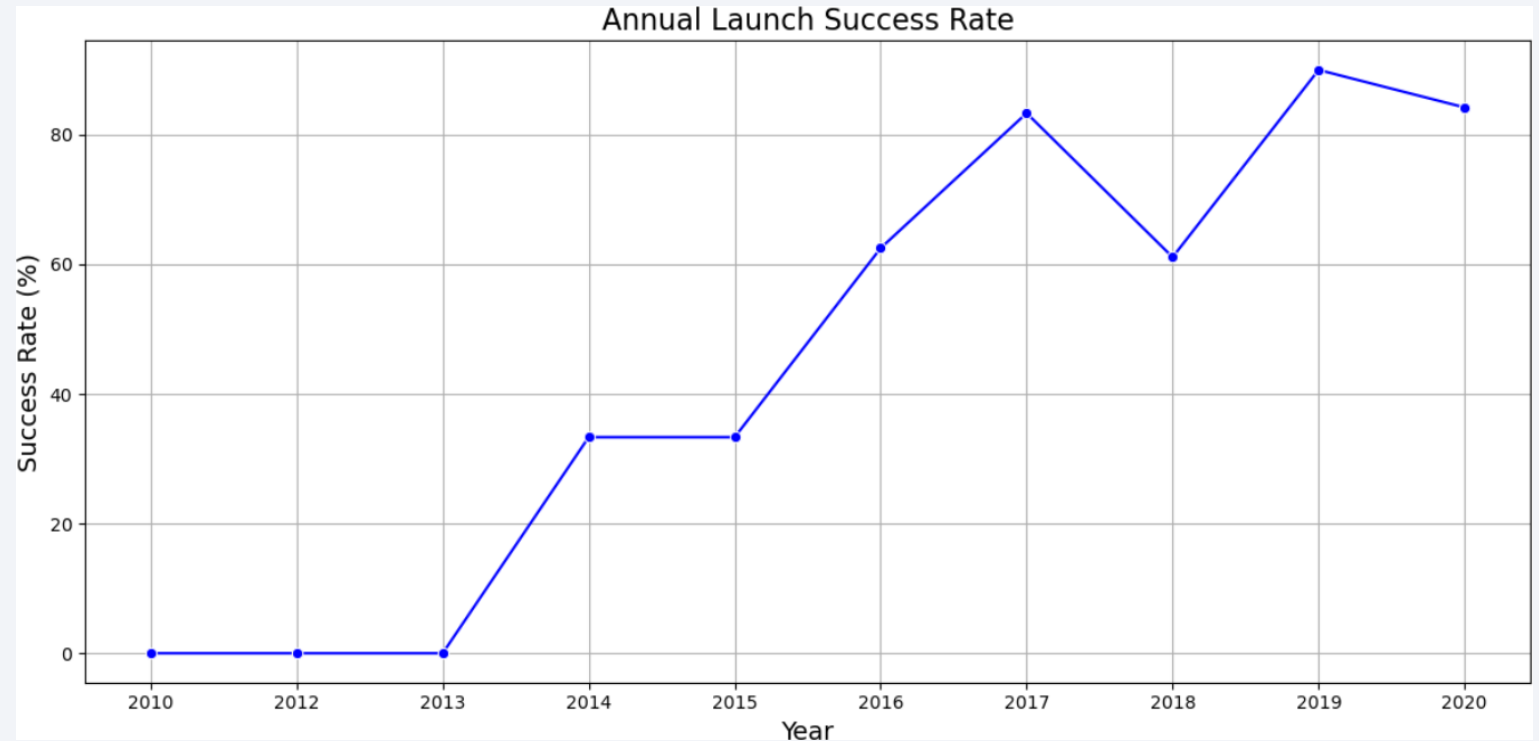
Payload vs. Orbit Type

- Below 6,000 kg, successes outnumber failures in nearly every orbit. GTO is the exception and stays mixed right across its range.
- Above 10,000 kg almost every point is VLEO, and most of those land. There are too few heavy launches to read much into that end of the plot.



Launch Success Yearly Trend

- Nothing lands before 2013. The rate jumps to 33% in 2014, holds flat through 2015, then climbs to 83% by 2017 and peaks near 90% in 2019.
- The dip to 61% in 2018 is the one break in the trend. Across the decade the rate runs from 0% to 84%, the largest single effect anywhere in this dataset.



All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

```
[21]: %sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db  
Done.
```

```
[21]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

- DISTINCT returns four site names. CCAFS LC-40 and CCAFS SLC-40 are the same Cape Canaveral pad under its old and current designations, so the data really covers three physical locations.

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
[26]: %sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db
```

Done.

```
[26]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- LIKE 'CCA%' with LIMIT 5 pulls the five earliest Cape Canaveral launches. All are Falcon 9 v1.0 and none attempted a powered landing — the first two came down by parachute and failed.

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[30]: %sql SELECT SUM("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[30]: SUM(PAYLOAD_MASS_KG_)
```

```
45596
```

- SUM over the rows where the customer is NASA (CRS) gives 45,596 kg delivered across the Commercial Resupply Services contract.

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
[34]: %sql SELECT AVG("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[34]: AVG(PAYLOAD_MASS_KG_)
```

```
2928.4
```

- AVG restricted to F9 v1.1 gives 2,928.4 kg, well under the Block 5 average and a fair measure of what that generation could lift.

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
[36]: %sql SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[36]: MIN(Date)
```

```
2015-12-22
```

- MIN over the ground pad successes returns 2015-12-22 — the ORBCOMM-2 mission, and SpaceX's first propulsive recovery on land.

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[38]: %sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_" < 6000
```

```
* sqlite:///my_data1.db  
Done.
```

```
[38]: Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

- Filtering on a drone ship success with a payload between 4,000 and 6,000 kg returns four boosters, all Full Thrust. Two of them (B1021.2 and B1031.2) are reflights, so a recovered booster can land a second time.

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

```
[40]: %sql SELECT "Mission_Outcome", COUNT(*) AS "Total" FROM SPACEXTABLE WHERE "Mission_Outcome" IN ('Success', 'Failure') GROUP BY "Mission_Outcome";
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[40]:
```

Mission_Outcome	Total
-----------------	-------

Success	98
---------	----

- Grouping on Mission_Outcome returns 98 successes and no rows matching 'Failure' exactly, because the single in-flight loss is stored as 'Failure (in flight)'. Mission success and landing success are different columns — most launches succeeded while losing the booster.

Boosters Carried Maximum Payload

Task 8

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

```
[42]: %sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTABLE);  
* sqlite:///my_data1.db  
Done.
```

```
[42]: Booster_Version
```

```
F9 B5 B1048.4
```

```
F9 B5 B1049.4
```

```
F9 B5 B1051.3
```

```
F9 B5 B1056.4
```

```
F9 B5 B1048.5
```

```
F9 B5 B1051.4
```

```
F9 B5 B1049.5
```

```
F9 B5 B1060.2
```

```
F9 B5 B1058.3
```

```
F9 B5 B1051.6
```

```
F9 B5 B1060.3
```

```
F9 B5 B1049.7
```

- The subquery finds the maximum payload and the outer query returns every booster that carried it. Twelve tie at 15,600 kg, all Block 5 and all reflights.

2015 Launch Records

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)= '2015' for year.

```
[69]: %sql
SELECT
CASE
  WHEN substr("Date", 6, 2) = '01' THEN 'January'
  WHEN substr("Date", 6, 2) = '02' THEN 'February'
  WHEN substr("Date", 6, 2) = '03' THEN 'March'
  WHEN substr("Date", 6, 2) = '04' THEN 'April'
  WHEN substr("Date", 6, 2) = '05' THEN 'May'
  WHEN substr("Date", 6, 2) = '06' THEN 'June'
  WHEN substr("Date", 6, 2) = '07' THEN 'July'
  WHEN substr("Date", 6, 2) = '08' THEN 'August'
  WHEN substr("Date", 6, 2) = '09' THEN 'September'
  WHEN substr("Date", 6, 2) = '10' THEN 'October'
  WHEN substr("Date", 6, 2) = '11' THEN 'November'
  WHEN substr("Date", 6, 2) = '12' THEN 'December'
  ELSE 'Unknown'
END AS "Month_Name",
"Mission_Outcome",
"Booster_Version",
"Launch_Site"
FROM
SPACEXTABLE
WHERE
substr("Date", 0, 5) = '2015';
```

```
* sqlite:///my_data1.db
Done.
```

```
[69]:
```

Month_Name	Mission_Outcome	Booster_Version	Launch_Site
January	Success	F9 v1.1 B1012	CCAFS LC-40
February	Success	F9 v1.1 B1013	CCAFS LC-40
March	Success	F9 v1.1 B1014	CCAFS LC-40
April	Success	F9 v1.1 B1015	CCAFS LC-40
April	Success	F9 v1.1 B1016	CCAFS LC-40
June	Failure (in flight)	F9 v1.1 B1018	CCAFS LC-40
December	Success	F9 FT B1019	CCAFS LC-40

- substr rebuilds the month name, which SQLite has no built-in function for. All seven 2015 launches flew from CCAFS LC-40, and the June flight on B1018 is the only in-flight failure in the dataset.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

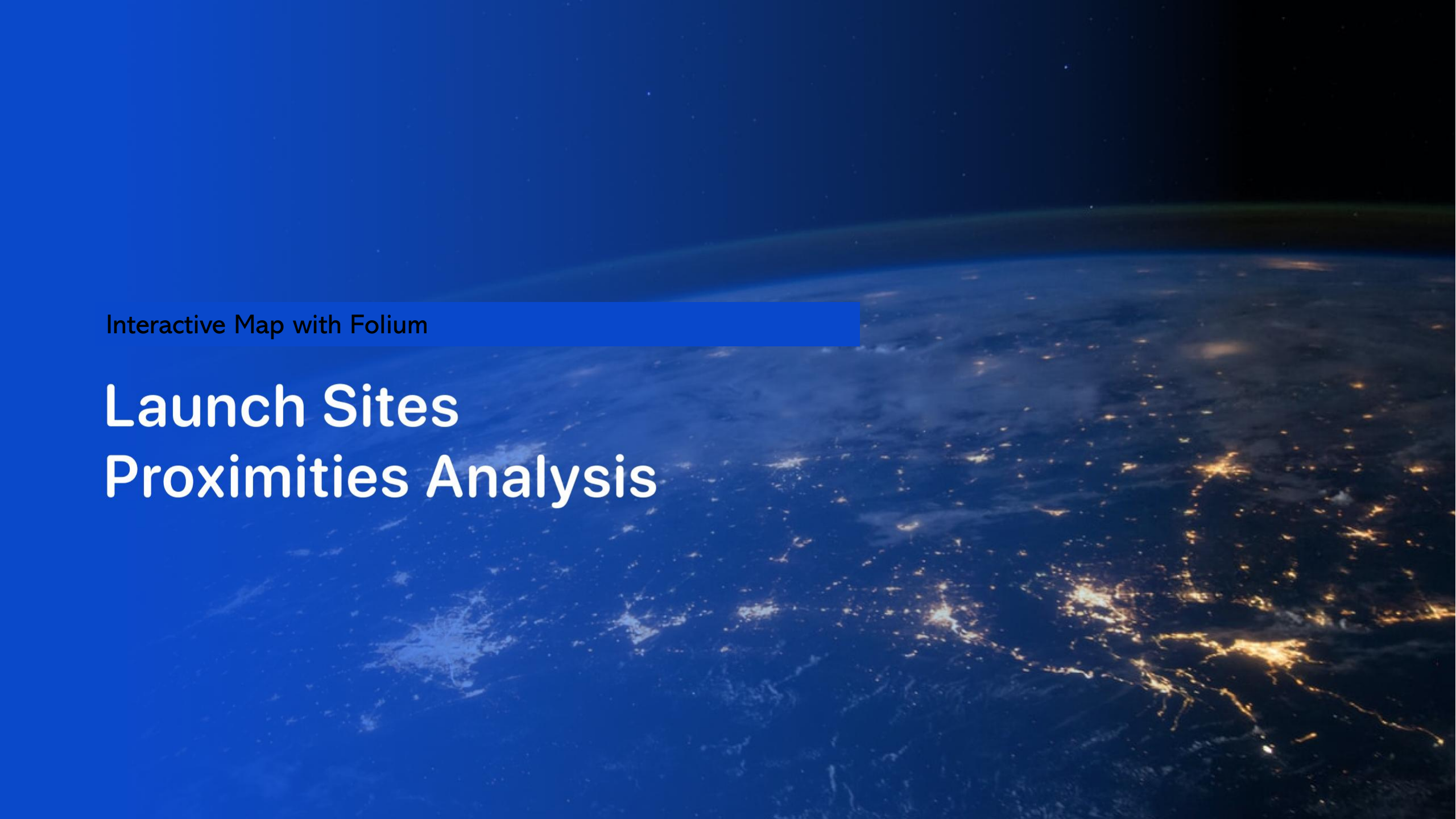
```
[81]: %%sql
SELECT
    "Landing_Outcome",
    COUNT(*) AS "Count"
FROM
    SPACEXTABLE
WHERE
    "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
    "Landing_Outcome"
ORDER BY
    COUNT(*) DESC;
```

```
* sqlite:///my_data1.db
Done.
```

```
[81]:
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- Across this window No attempt leads with 10, and drone ship successes and failures tie at 5 each. Recovery was still close to a coin flip before 2017.

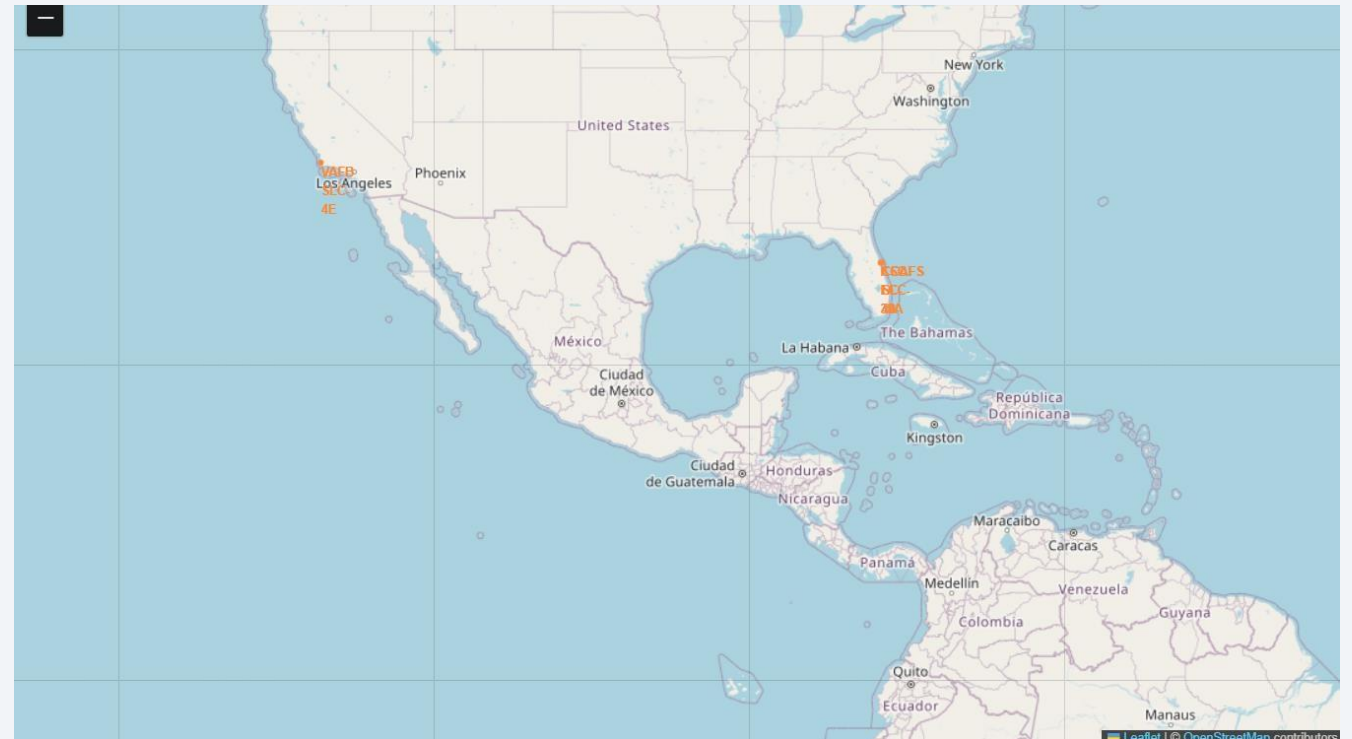


Interactive Map with Folium

Launch Sites Proximities Analysis

All Launch Sites on a Global Map

- All four pads sit on a United States coastline.
 - Three cluster on the Florida east coast near 28.5° N: CCAFS LC-40, CCAFS SLC-40 and KSC LC-39A.
 - VAFB SLC-4E is at 34.63° N on the California coast, the furthest of the four from the Equator.
- Low latitude buys some of Earth's rotational velocity, which cuts the delta-v needed to reach orbit. Coastal siting keeps failed launches and spent stages out over water.



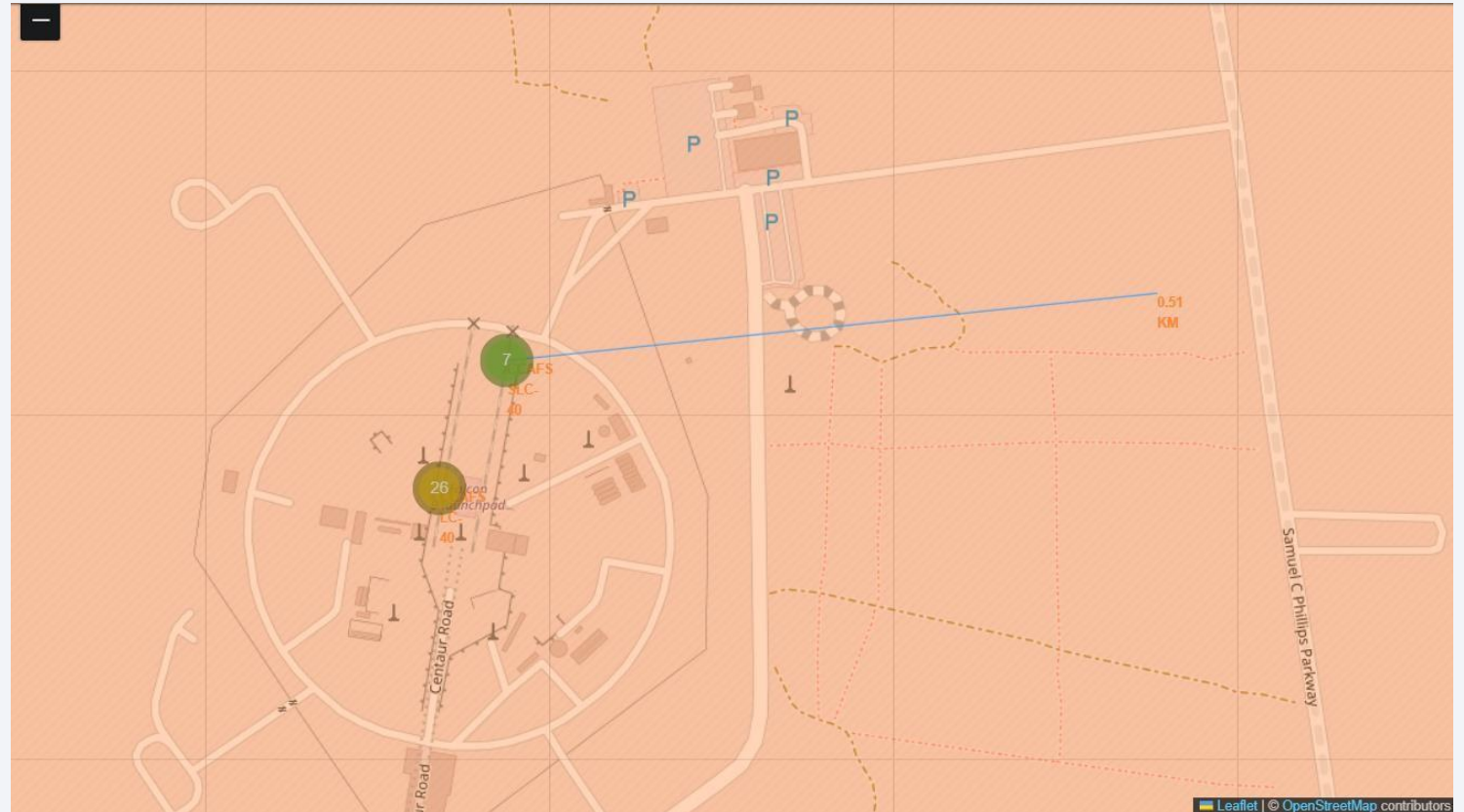
Launch Outcomes Marked by Site

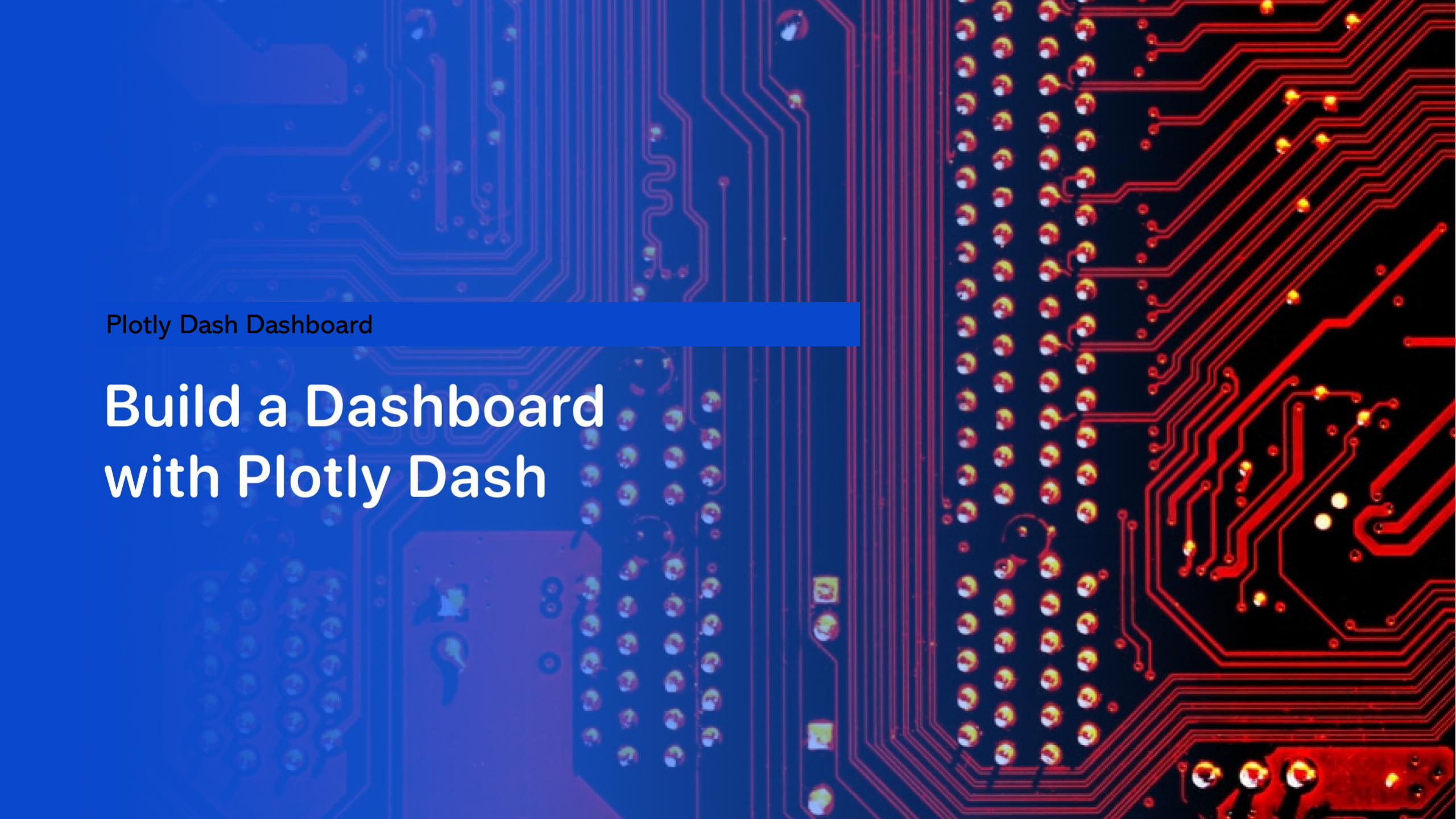
- Green markers are successful landings, red are failures, grouped in a MarkerCluster because many launches share identical coordinates.
- Expanding the CCAFS LC-40 cluster spreads 26 launches into 19 red markers and 7 green, so this pad carried most of the early attempts and most of the early losses.
- Without clustering every launch from one pad stacks on the same point and the site looks like a single marker.



Distance from CCAFS SLC-40 to Its Proximities

- The PolyLine runs from CCAFS SLC-40 to the nearest coastline, 0.51 km away, measured with the Haversine formula.
- Half a kilometre is close enough that a failed ascent clears land almost immediately, which is the reason pads are sited on a coast in the first place.
- The same method gives the distances to the nearest railway, highway and city.



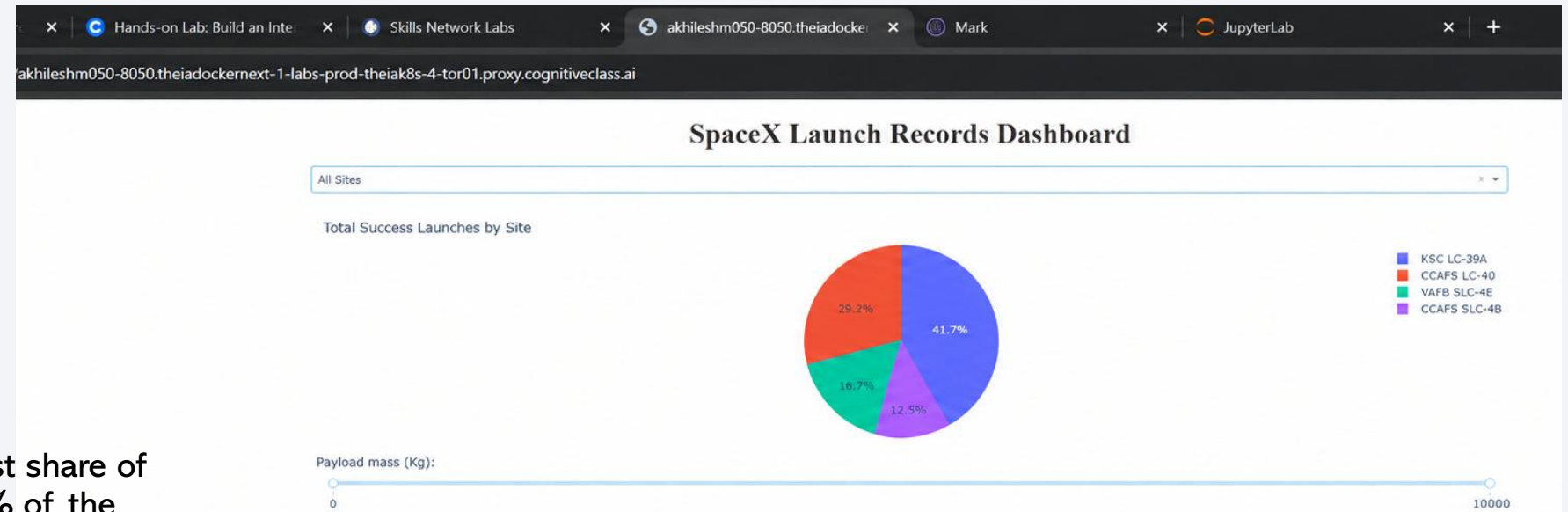
The background of the slide is a close-up photograph of a circuit board. The left side is blue-tinted, showing various components and traces. The right side is red-tinted, showing a dense array of gold-plated pins or components. The text is overlaid on the blue section.

Plotly Dash Dashboard

Build a Dashboard with Plotly Dash

Launch Success Count for All Sites

- KSC LC-39A: 41.7%
 - CCAFS LC-40: 29.2%
 - VAFB SLC-4E: 16.7%
 - CCAFS SLC-40: 12.5%
- KSC LC-39A holds the largest share of successful landings at 41.7% of the total.
 - This is a count of successes, not a success rate — a pad that flies more often will win on count alone.



Launch Site with the Highest Success Ratio

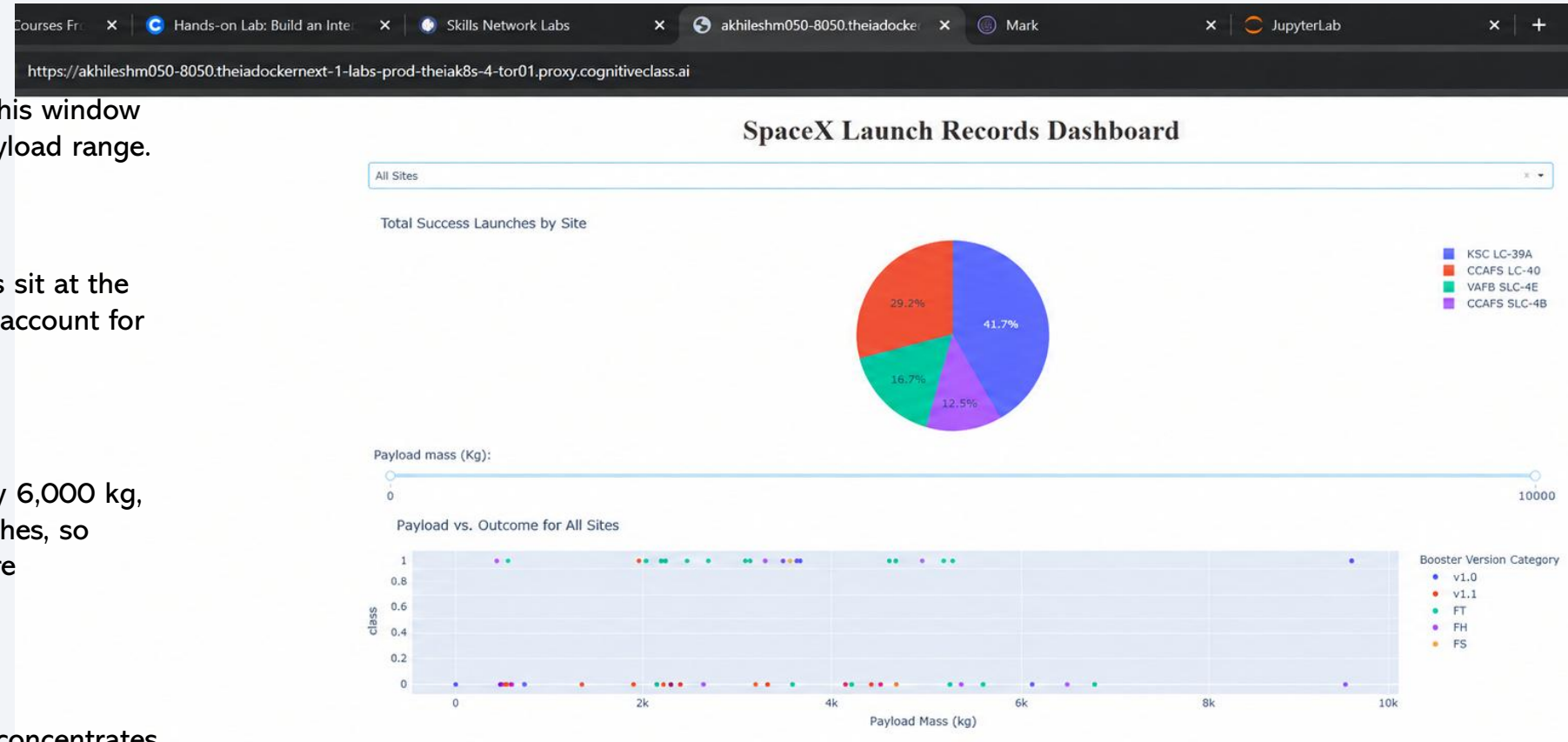
Total Success Launches for site KSC LC-39A



- KSC LC-39A — Class 1 (landed): 76.9%
- KSC LC-39A — Class 0 (lost): 23.1%
- Selecting KSC LC-39A in the dropdown switches the pie from share-of-successes to that pad's own outcome split. 76.9% is the best ratio of the four.
- KSC LC-39A also came into service late, so its record runs on mature boosters. Site and era are hard to separate here.

Payload vs. Launch Outcome for All Sites

- FT is the most-flown booster in this window and lands across most of the payload range.
- The early v1.0 and v1.1 boosters sit at the low end of the payload axis and account for much of the class 0 row.
- Successes thin out above roughly 6,000 kg, but so does the number of launches, so payload mass and sample size are confounded at that end.
- Sliding the payload range down concentrates the successes below about 6,000 kg.



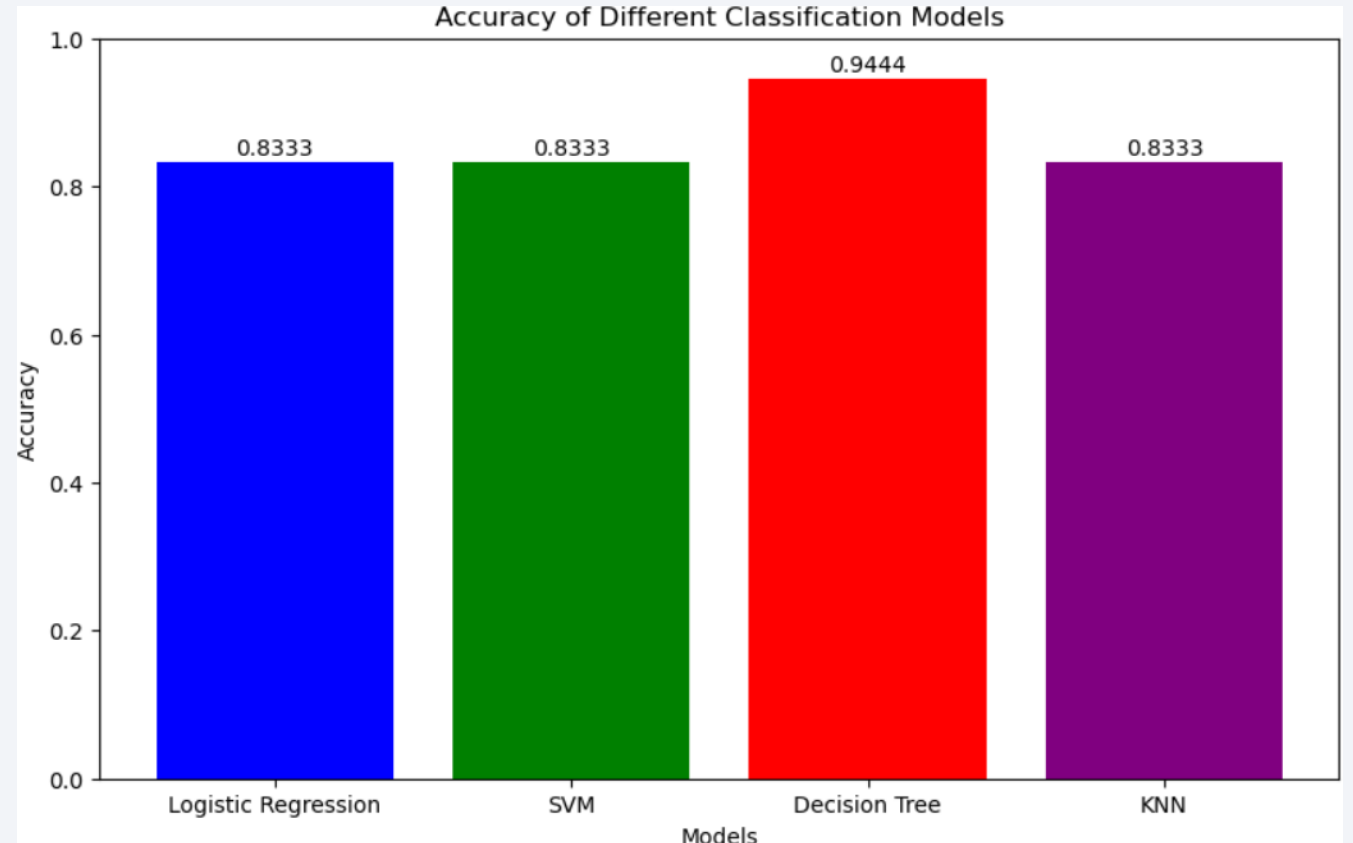


Predictive Analysis

Predictive Analysis (Classification)

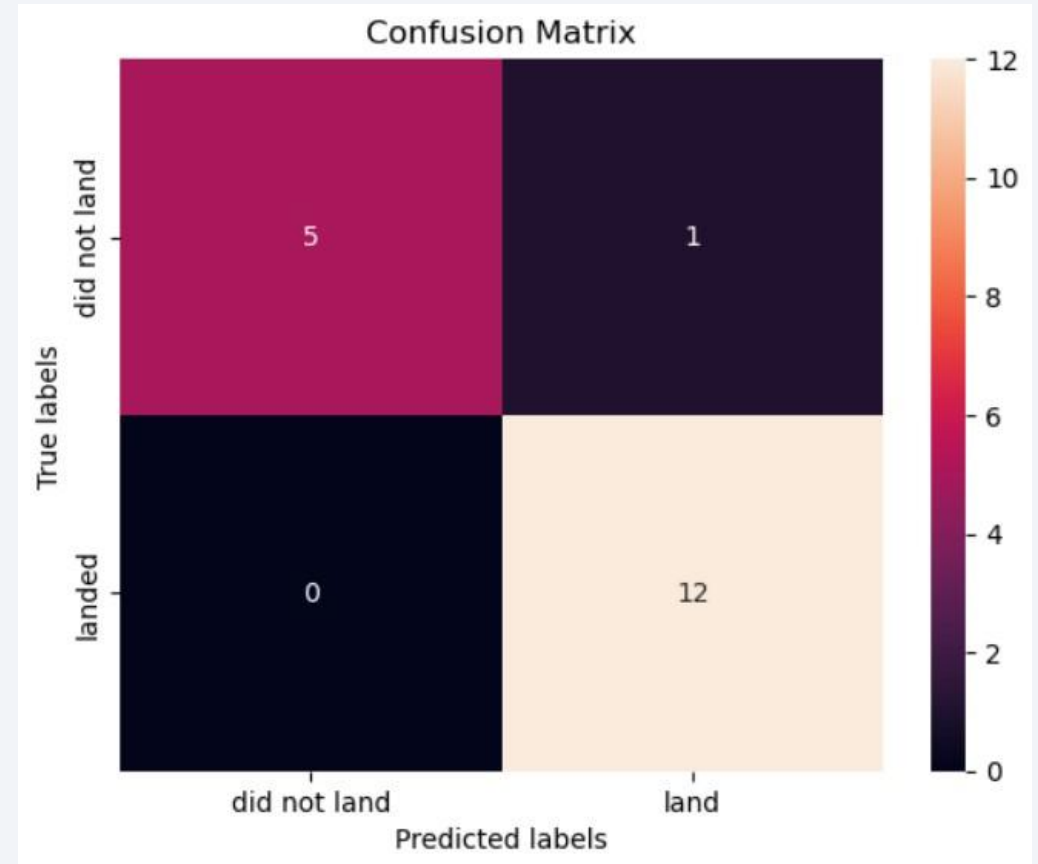
Classification Accuracy

- **Decision tree: 0.9444**
- **Logistic regression: 0.8333**
- **SVM: 0.8333**
- **KNN: 0.8333**
- In counts, the tree got 17 of 18 launches right and the other three got 15.
- On a test set this small, one launch is the entire difference. The tree is the pick here rather than a proven best.



Confusion Matrix

- **17 of 18 correct.**
 - 12 true positives, 5 true negatives, one miss, 94.44% accuracy on the held-out launches.
- **No false negatives.**
 - Every launch that actually landed was predicted to land, so recall on the positive class is 100%.
- **One false positive.**
 - The model predicted a landing on a launch that was lost. For cost estimation that is the direction that hurts, because it understates what the launch really cost.
- **What this does not prove.**
 - 18 test launches and a roughly 2:1 class imbalance leave wide error bars. The matrix shows a model that leans optimistic about recovery.



Conclusions

- Flight number is the strongest single predictor in the dataset. Landing success goes from 0% before 2013 to 84% in 2020, and every orbit holding a perfect record was flown mostly in those later years.
- KSC LC-39A has both the largest share of successful landings (41.7%) and the highest success ratio (76.9%), but it opened later than the Cape pads. The site advantage and the hardware advantage cannot be pulled apart from this data.
- Payload mass matters, but not on its own. Successes concentrate below 6,000 kg, and the heaviest flights are also the most recent, so mass, orbit and era are tangled together.
- Launch sites are chosen for safety and logistics, not performance: all four sit within about half a kilometre of a coastline and close to rail and road, while staying clear of towns.
- The decision tree predicts a landing correctly 17 times out of 18 and never misses a real landing, which is enough to be useful for estimating what a Falcon 9 launch costs. It is trained on 90 launches and tested on 18, so the next thing this work needs is more data, not a better model.

Appendix

- All notebooks, the Dash app and the cleaned datasets:
- <https://github.com/akhileshm0507/IBM-Data-Science-Capstone-Project>
- **Also held in the repository**
 - Flight Number vs. Payload Mass scatter plot, the one EDA chart not shown in the results section.
 - GridSearchCV best_params_ and best_score_ for all four classifiers.
 - Confusion matrices for logistic regression, SVM and KNN alongside the decision tree.
 - The one-hot encoded feature table used for model training.
- **Limitations worth stating**
 - $n = 90$ launches, with an 18-launch test set and a roughly 2:1 class imbalance.
 - The data ends in 2020, so Block 5 reliability is under-represented.

Thank you!

